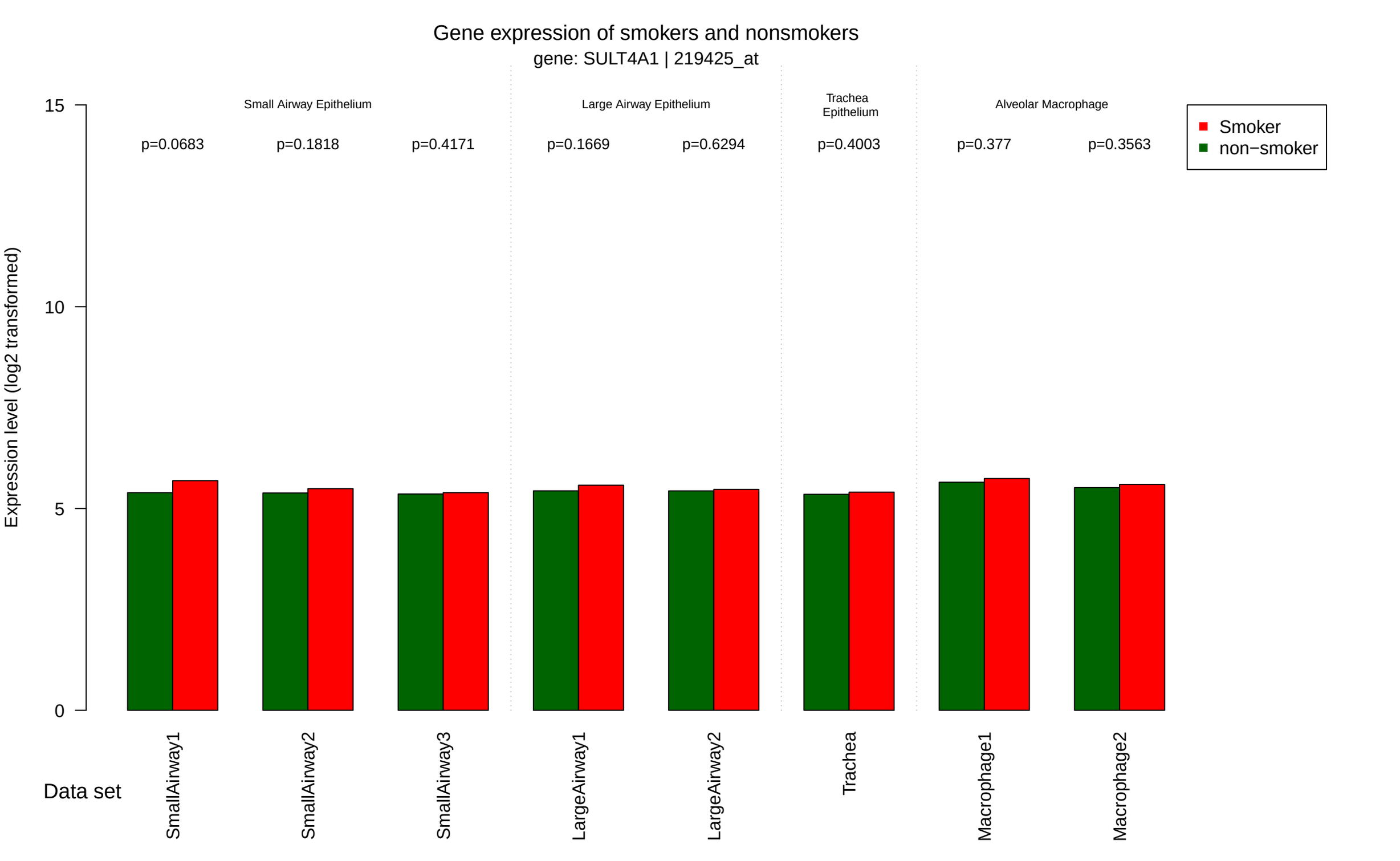
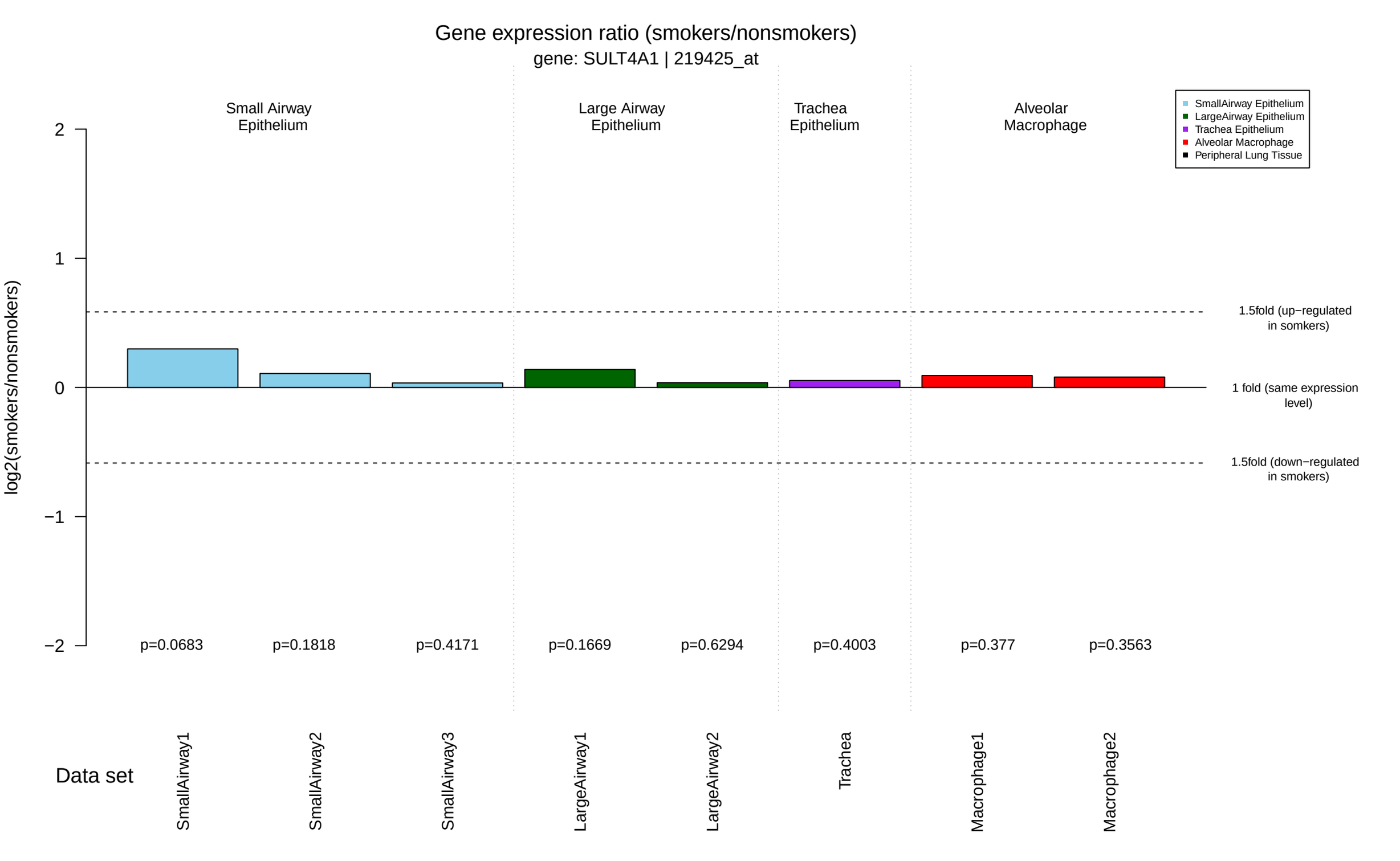
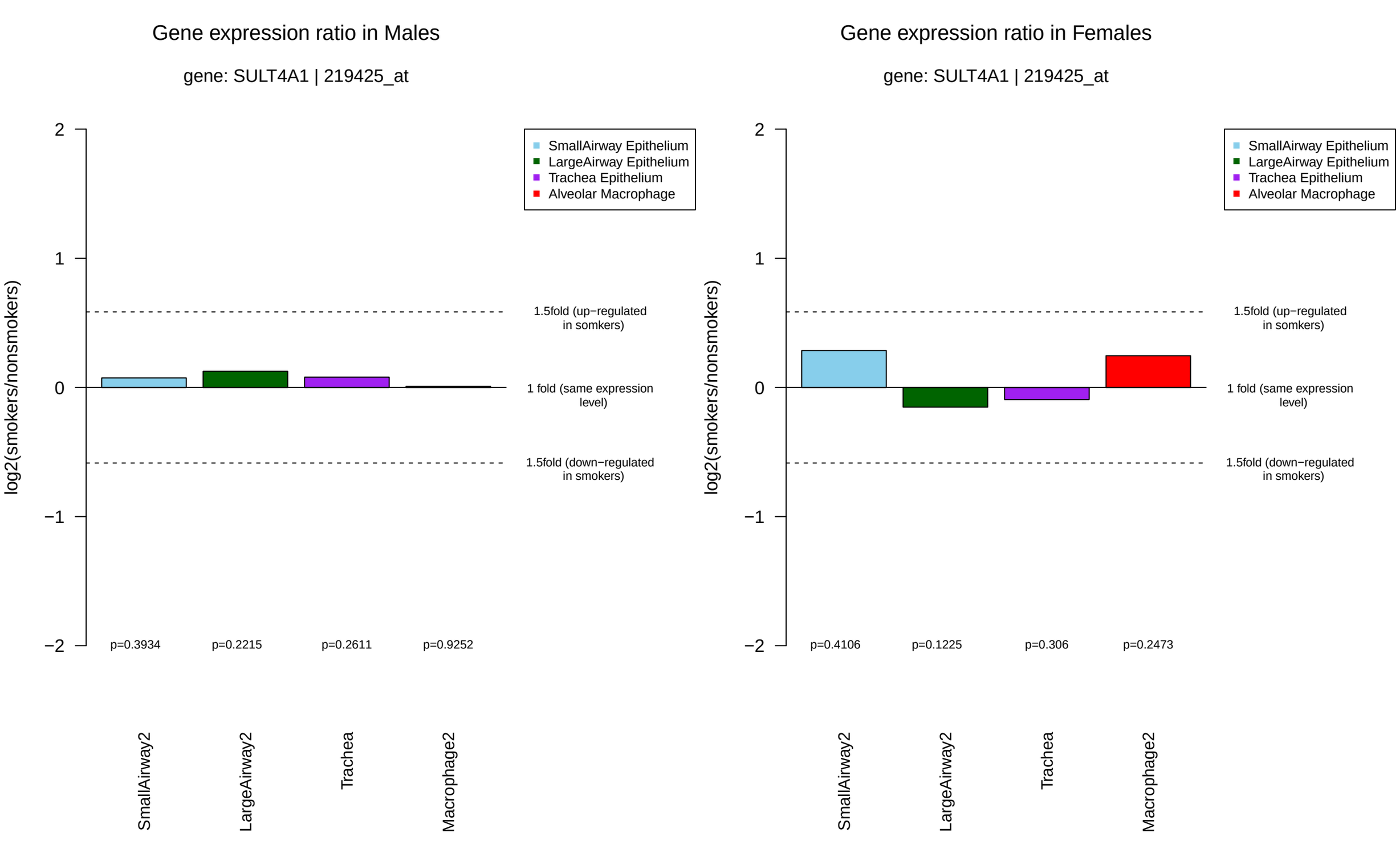
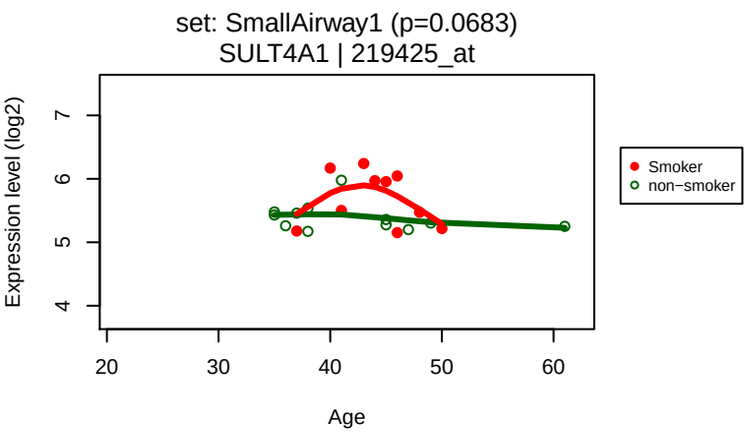
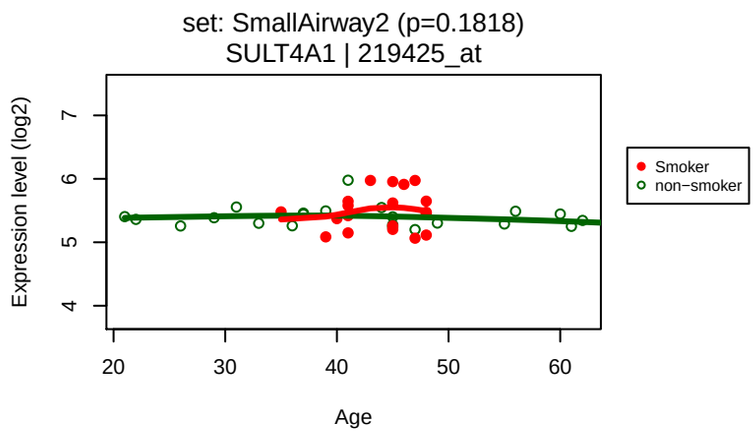
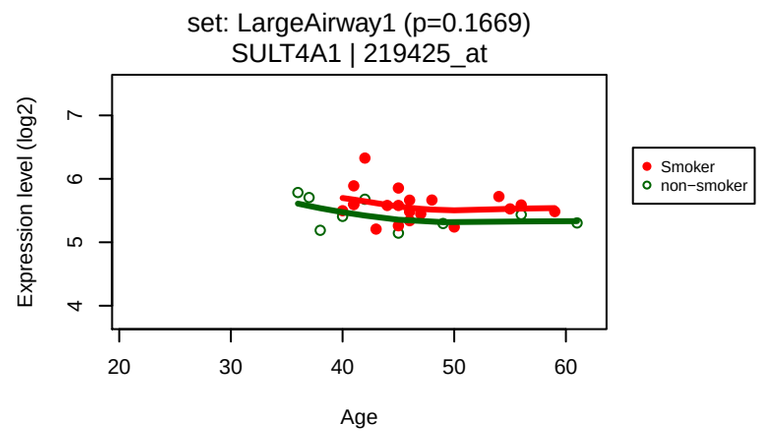

Gene expression of smokers and nonsmokers
gene: SULT4A1 | 219425_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0683
p=0.1818
p=0.4171
p=0.1669
p=0.6294
p=0.4003
p=0.377
p=0.3563
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

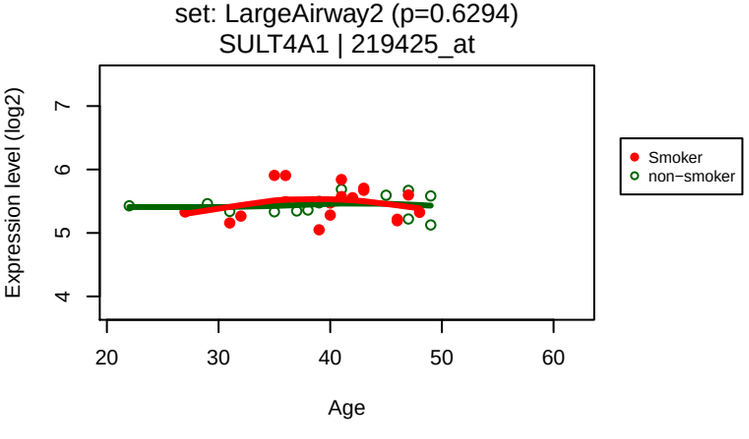

Gene expression ratio (smokers/nonsmokers)
gene: SULT4A1 | 219425_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0683
p=0.1818
p=0.4171
p=0.1669
p=0.6294
p=0.4003
p=0.377
p=0.3563
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

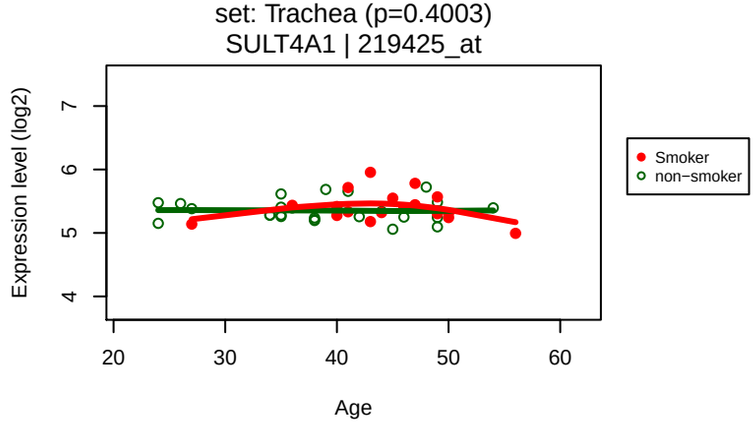

Gene expression ratio in Males
gene: SULT4A1 | 219425_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3934
p=0.2215
p=0.2611
p=0.9252
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SULT4A1 | 219425_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4106
p=0.1225
p=0.306
p=0.2473
SmallAirway2
LargeAirway2
Trachea
Macrophage2

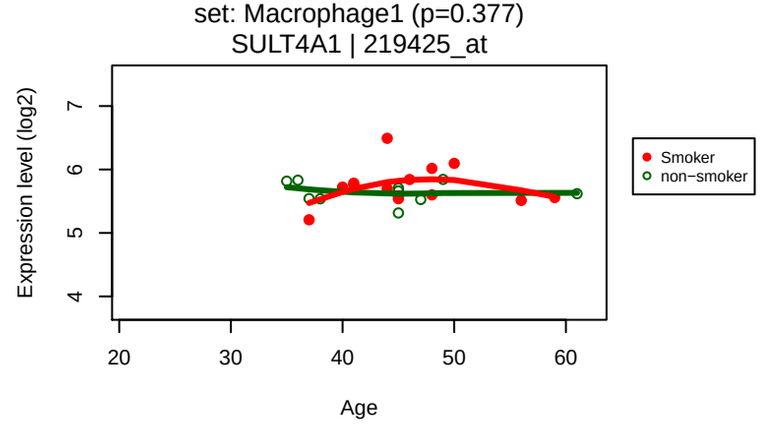

set: SmallAirway1 (p=0.0683)
SULT4A1 | 219425_at
Expression level (log2)
Age
Smoker
non−smoker

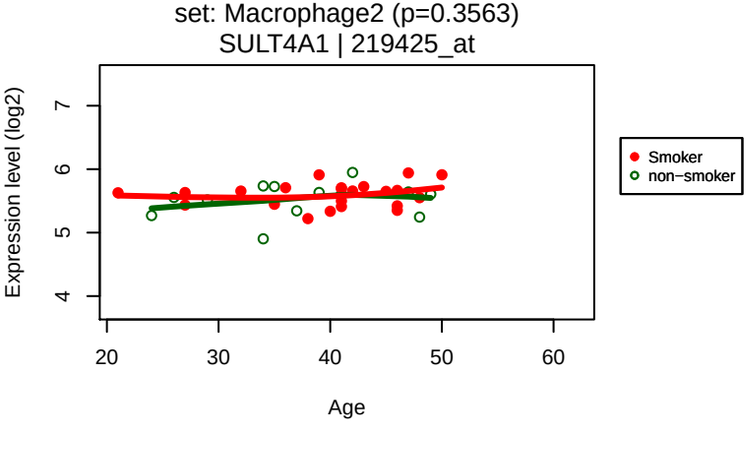

set: SmallAirway2 (p=0.1818)
SULT4A1 | 219425_at
Expression level (log2)
Age
Smoker
non−smoker

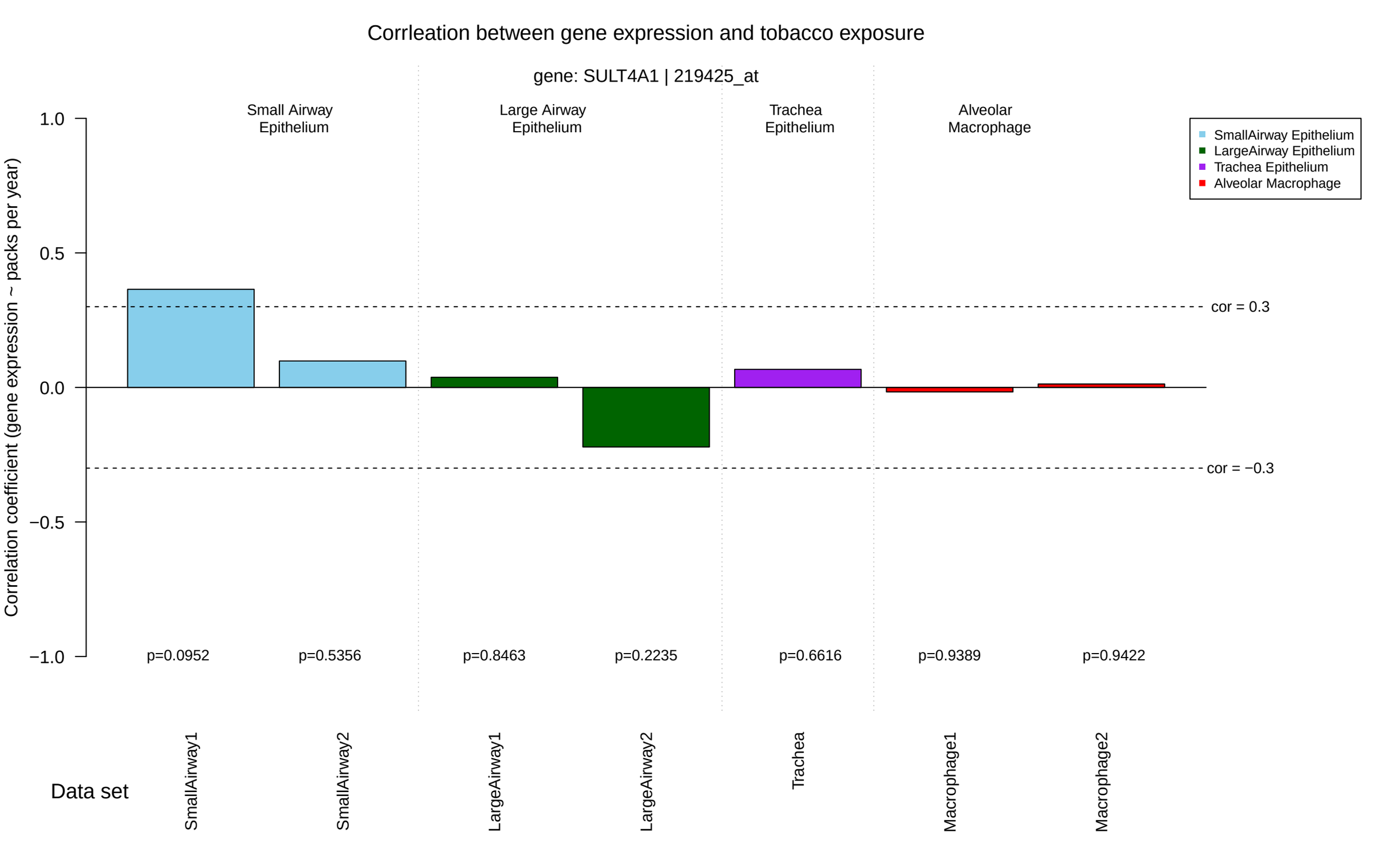

set: LargeAirway1 (p=0.1669)
SULT4A1 | 219425_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6294)
SULT4A1 | 219425_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4003)
SULT4A1 | 219425_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.377)
SULT4A1 | 219425_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3563)
SULT4A1 | 219425_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SULT4A1 | 219425_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0952
p=0.5356
p=0.8463
p=0.2235
p=0.6616
p=0.9389
p=0.9422
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2